# TEST RESULTS FOR DEMAND FORECASTING AND INVENTORY OPTIMIZATION

1) Lags , Features, Rolling mean

```
--- Feature Engineering Complete! ---
Processed Shape: (5832737, 23)
              date          item_id store_id  sales  sell_price  lag_1  \
5832734 2016-05-20  HOUSEHOLD_2_516     CA_1      1        5.94    0.0
5832735 2016-05-21  HOUSEHOLD_2_516     CA_1      0        5.94    1.0
5832736 2016-05-22  HOUSEHOLD_2_516     CA_1      0        5.94    0.0

         rolling_mean_7
5832734        0.000000
5832735        0.142857
5832736        0.142857
```

2) Naive Baseline

```
--- Global Naive Baseline (using sales_long) -
RMSE: 2.7611 | MAE: 1.3477
```

3) Prophet Baseline

```
--- Training Prophet Baseline ---
Prophet Validation RMSE (for FOODS_1_001 at CA_1): 1.3705
```

4) LightGBM

```
--- Training LightGBM ---
[LightGBM] [Warning] Categorical features with more bins than the configured maximum bin number found.
[LightGBM] [Warning] For categorical features, max_bin and max_bin_by_feature may be ignored with a large number of c
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of testing was 0.075852 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4685
[LightGBM] [Info] Number of data points in the train set: 5747365, number of used features: 18
[LightGBM] [Info] Start training from score 1.323561
Training until validation scores don't improve for 50 rounds
[50]    valid_0's rmse: 2.037
[100]   valid_0's rmse: 2.01062
[150]   valid_0's rmse: 2.00783
[200]   valid_0's rmse: 2.00553
[250]   valid_0's rmse: 2.0046
[300]   valid_0's rmse: 2.00397
[350]   valid_0's rmse: 2.00231
[400]   valid_0's rmse: 2.00213
[450]   valid_0's rmse: 2.00116
[500]   valid_0's rmse: 2.00039
Did not meet early stopping. Best iteration is:
[500]   valid_0's rmse: 2.00039

LightGBM Validation RMSE: 2.0004
LightGBM Validation MAE:  1.0780
```

5) Hierarchical Aggregation

```
--- Executing Bottom-Up Hierarchical Aggregation ---
Hierarchical aggregation successfully computed!
Item-Store Rows: 5,832,737
Store-Level Rows: 1,913
State-Level Rows: 1,913
```

6) OR Tools Inventory Optimization

```
--- Setting up OR-Tools Inventory Optimization ---

--- Optimization Complete: Optimal Solution Found ---
Optimal Order Quantity : 0.00 units
Ending Inventory       : 90.00 units
Expected Stockout (Shortage): 0.00 units
Minimum Total Cost     : $45.00
```

7) Temporal Fusion Transformer

```
Trainable params: 67.4 K
Non-trainable params: 0
Total params: 67.4 K
Total estimated model params size (MB): 0.270
Modules in train mode: 258
Modules in eval mode: 0
Total FLOPs: 0
Epoch 1/1 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 92327/92327 0:31:09 • 0:00:00 48.95it/s v_num: 0.000 train_loss_step:
                                                                                  0.824 val_loss: 1.672
                                                                                  train_loss_epoch: 0.715
/usr/local/lib/python3.13/dist-packages/lightning/pytorch/utilities/_pytree.py:21: `isinstance(treespec, LeafSpec)`
is deprecated, use `isinstance(treespec, TreeSpec) and treespec.is_leaf()` instead.
INFO: `Trainer.fit` stopped: `max_epochs=2` reached.
INFO:lightning.pytorch.utilities.rank_zero:`Trainer.fit` stopped: `max_epochs=2` reached.
```